

Math 55 quiz 04/28/2026

You must type or neatly handwrite each answer on a separate piece of paper and select the pages for each question when you upload to Gradescope. Feel free to use more than one sheet for an answer if needed. Note that both PDF and picture/photo format (JPG) are accepted. You must show your work. During the quiz, you may consult the textbook, your own notes and other course materials you yourself prepared before the beginning of the quiz. You may not consult resources online. You can post a private question to Ed Discussion to instructors only, but do not make public posts on Ed Discussion, Discord, or other online forums.

By submitting this quiz, you agree to adhere to the statement below: “As a member of the UC Berkeley community, I act with honesty, integrity, and respect for others. All of the work submitted here is mine, and I have not discussed the content of this exam with any other person. I have not consulted any resources beyond the course materials and my notes during the exam. I have not used calculators, computers, or the internet to do calculations or look up answers.”

1. Construct a random graph as follows: take n disconnected vertices, and, for every pair of vertices x_1, x_2 , add the edge (x_1, x_2) to the graph at random with probability $1/2$. What is the expected number of 3-cycles in this graph? Provide a rigorous proof.
2. In some tabletop games, there is the concept of an “exploding die.” Say I roll a twenty-sided die. If I get a 20, I roll again, adding the resulting number to the original roll. If I get a 20 on the next roll, I roll again, and so on. Let S be the sum of all the numbers I have rolled when the process ends. Use indicator variables and linearity of expectation to prove that $E[S] = c + dE[S]$ for some constants c and d which you should find, and hence determine $E[S]$.